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Project Description:

The Smart Electric Scooter Sharing System will mirror real-life scooter sharing activities such as booking, riding, returning, billing, and fleet upkeep. The database will serve as the backbone for a mobile or web application that allows users to locate, rent, and ride electric scooters within designated zones.

Main Features of the System

1.User Management:

The database will maintain records of all users, including UserID, Name, Phone Number, Email, Password Hash, Registration Date, and Verification Status. This will support user authentication and profile management.

2.Scooter Fleet Management:

The system will track detailed information about each scooter, including ScooterID, Model,Battery Level, Current Status (available, in use, maintenance, low battery), Last Known GPS Coordinates, and Timestamp of Last Update. This enables real-time tracking and fleet monitoring.

3.Location/Zones Management:

The database will store information about parking locations, including LocationID, Zone Name, GPS Coordinates, and Maximum Parking Capacity. This helps users find available scooters and parking spots.

4. Trip Recording:

Every ride will be recorded as a trip with attributes including TripID, UserID, ScooterID, Start Location ID, EndLocation ID, Start Time, End Time, Distance in km, Cost, and Trip Status (ongoing, completed, cancelled). This forms the core transactional data.

5. Payment Processing:

The system will handle all financial transactions through a Payment table with PaymentID, TripID (the trip being paid for), Amount, Payment Method (card, wallet, promo), Payment Time, and Payment Status.

6.Ratings and Feedback:

•Users can rate their riding experience through a Rating table containingRating|D, UserID, TripID, ScooterID, Score (1-5), Comment, and Rating Time. This provides valuable feedback for service improvement.

7. Staff Management:

The database will track staff members including StaffID, Name, Role(mechanic, support, manager), Shift Schedule, and Phone Number.

8. Maintenance Scheduling:

A Maintenance table will record all scooter service activities withMaintenanceID, ScooterID, Request Time, Completion Time, IssueDescription, Maintenance Status, and Technician ID (referencing Staff.

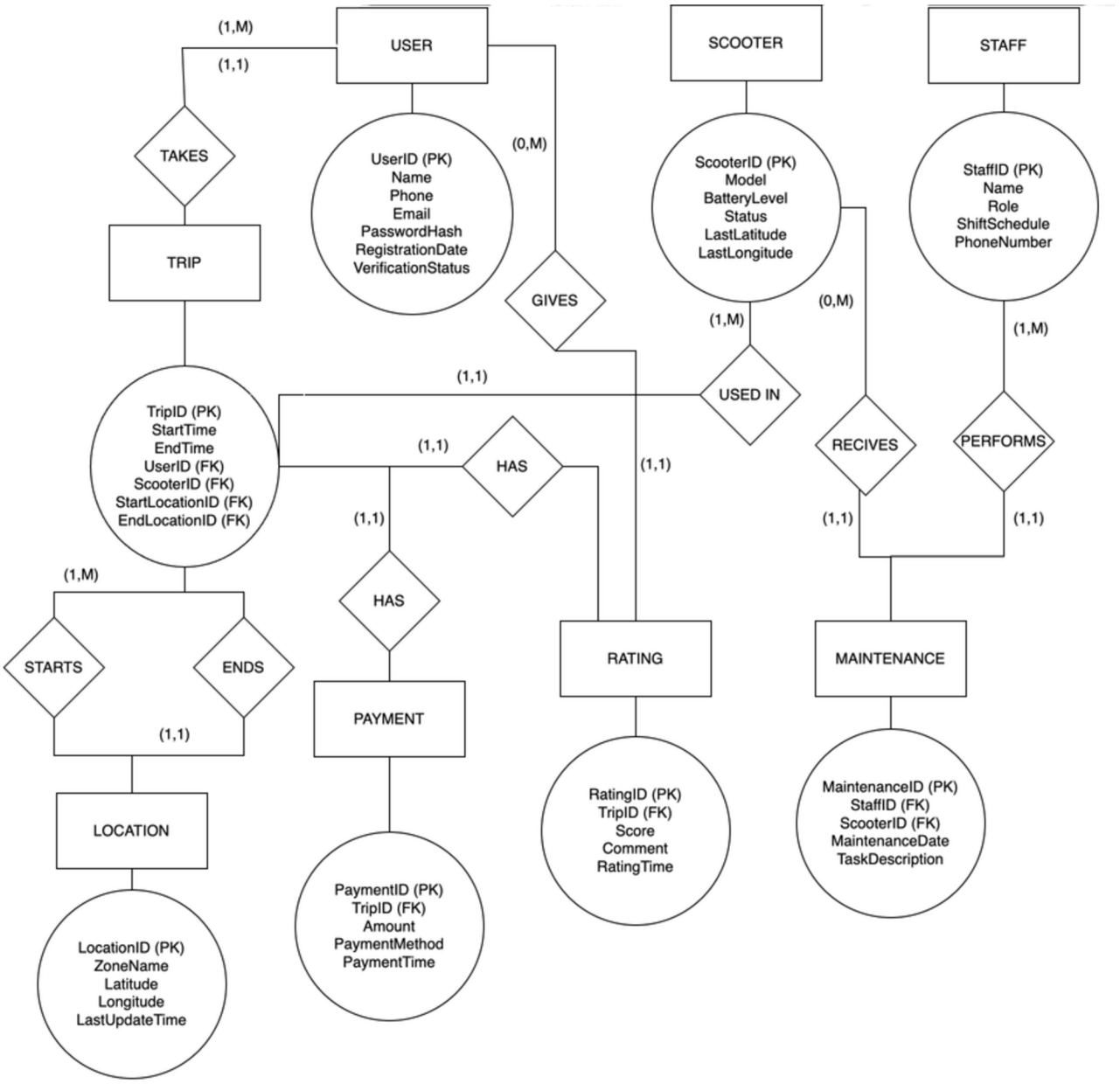
GLOSSARY TABLE

TERM	DESCRIPTION	RELATED TERMS
User	A registered person who uses the scooter sharing system to rent and ride scooters.	Trip, Rating, Payment
Scooter	An electric scooter available for users to rent within the system.	Trip, Maintenance, Rating
Trip	A ride taken by a user using a scooter from a start location to an end location.	User, Scooter, Payment, Location
Payment	A financial transaction made by the user to pay for a completed trip.	Trip, User
Rating	Feedback given by a user after completing a trip, usually between 1–5 stars.	User, Trip, Scooter
Verification Status	The confirmation status of a user's account after registration.	User

Location	A designated parking or operational zone where scooters can be parked or picked up.	Trip, Scooter
Staff	Employees responsible for managing scooters and performing maintenance tasks.	Maintenance
Maintenance	Service or repair activity performed on scooters to keep them operational.	Scooter, Staff
Battery Level	The current charge percentage of a scooter's battery.	Scooter
Payment Method	The method used to pay for a trip (card, wallet, etc.).	Payment
Distance	The total kilometers traveled during a trip.	Trip
Trip Status	The current state of a trip (ongoing, completed, cancelled).	Trip
Technician	A staff member responsible for repairing or servicing scooters.	Staff, Maintenance
Zone Capacity	The maximum number of scooters allowed in a specific location zone.	Location, Scooter

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ENTRY RELATIONSHIP DIAGRAM



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Table of Data Volumes and Operations

Entity / Table	Expected Data Volume	Common Operations
User	5,000	Register new user, update profile, verify account, view trip history
Scooter	200	Add new scooter, update battery level, change scooter status, track location
Location	50	Add parking zone, update zone capacity, view nearby scooter locations
Trip	50,000	Start trip, end trip, calculate trip cost, view active trips
Payment	50,000	Process payment, update payment status, issue refund, view revenue
Rating	25,000	Submit rating, view scooter feedback, calculate average rating
Staff	20	Add staff member, assign roles, update work schedule
Maintenance	500	Record maintenance task, assign technician, update maintenance status, view maintenance history

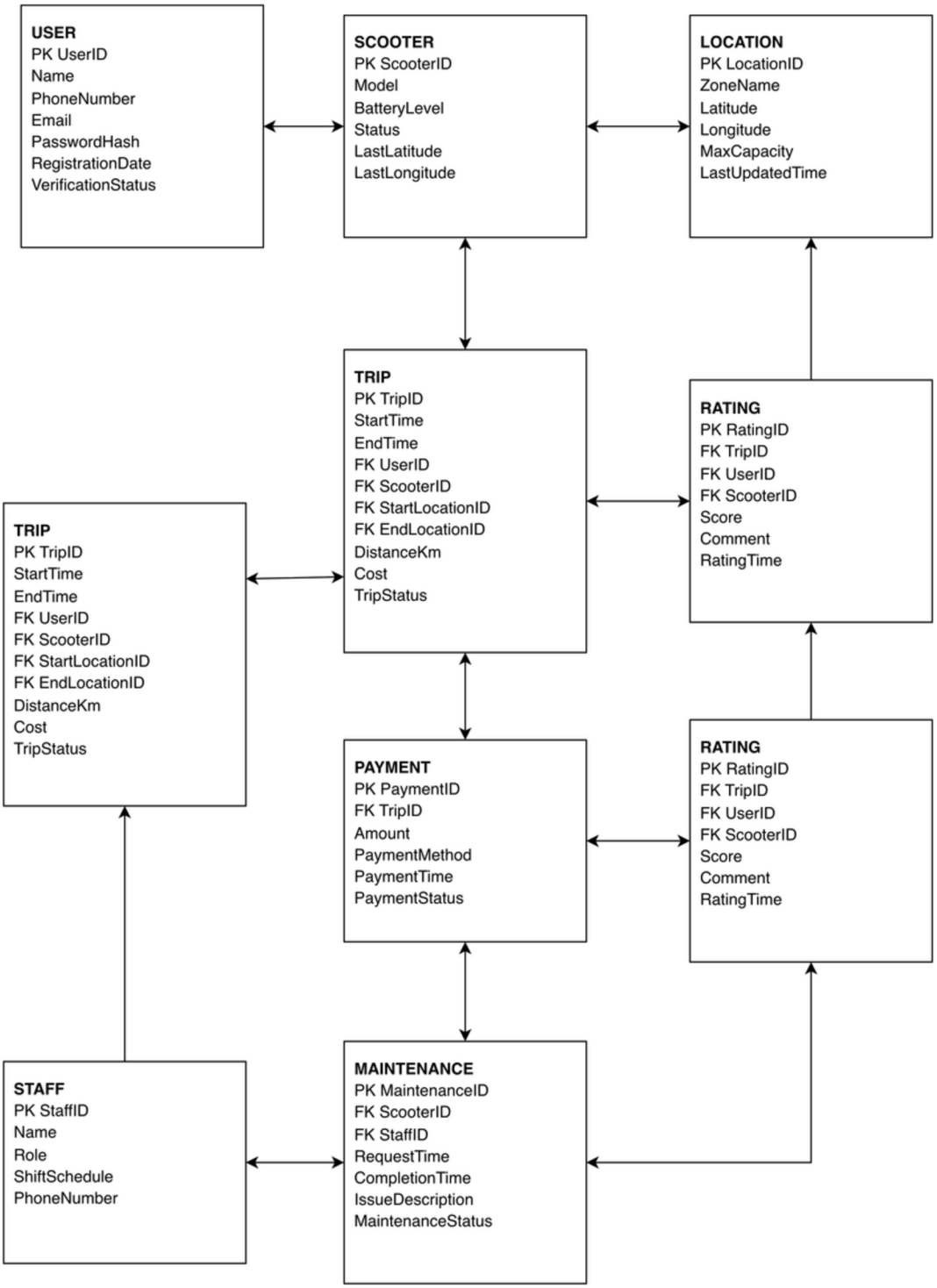
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Main Operations of Smart Electric Scooter Sharing System

OPERATION	DESCRIPTION
Register User	Allows a new user to create an account by providing personal details such as name, phone number, email, and password.
Login User	Authenticates registered users so they can access the scooter sharing system.
View Available Scooters	Displays nearby scooters that are currently available for rental.
Start Trip	Allows a user to unlock and start riding a selected scooter. The system records trip start time and location.
End Trip	Ends the ride when the user parks the scooter in a valid zone. The system records end time and location.
Calculate Trip Cost	Calculates the total trip cost based on ride duration or distance travelled.
Process Payment	Handles payment for the completed trip using available payment methods such as card or digital wallet.
Submit Rating	Allows users to provide feedback and rate their riding experience after completing a trip.
Update Scooter Status	Updates scooter availability status such as available, in use, or under maintenance.
Track Scooter Location	Monitors scooter GPS location to help users find nearby scooters and for fleet management.
Schedule Maintenance	Records maintenance requests and assigns technicians to repair scooters.
Manage Parking Zones	Allows administrators to add or update scooter parking zones and their capacities.

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LOGICAL SCHEMA (RELATIONAL MODEL)



Attributes and Foreign Keys Table

TABLE	ATTRIBUTES	FOREIGN KEYS
Users	user_id (PK), name, phone, email, password	-
Scooters	scooter_id (PK), model, battery_level, status, location	-
Trips	trip_id (PK), user_id, scooter_id, start_time, end_time, distance, cost	user_id → Users, scooter_id → Scooters
Payments	payment_id (PK), trip_id, amount, payment_method, payment_time	trip_id → Trips
Ratings	rating_id (PK), trip_id, rating, comment	trip_id → Trips
Maintenance	maintenance_id (PK), scooter_id, technician_id, issue_description, status	scooter_id → Scooters, technician_id → Staff
Staff	staff_id (PK), name, role, shift_schedule	-
Location	zone_id (PK), zone_name, gps_coordinates, capacity	-

SQL Code to Create Tables

```sql
CREATE TABLE Users (
user_id INT PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(100),
phone VARCHAR(20),
email VARCHAR(100),
password VARCHAR(255)
);

CREATE TABLE Scooters (
scooter_id INT PRIMARY KEY AUTO_INCREMENT,
model VARCHAR(50),
battery_level INT,
status VARCHAR(20),
location VARCHAR(100)
);

CREATE TABLE Trips (
trip_id INT PRIMARY KEY AUTO_INCREMENT,
user_id INT,
scooter_id INT,
start_time DATETIME,
end_time DATETIME,
start_location VARCHAR(100),
end_location VARCHAR(100),
distance FLOAT,
cost DECIMAL(10,2),
FOREIGN KEY (user_id) REFERENCES Users(user_id),
FOREIGN KEY (scooter_id) REFERENCES Scooters(scooter_id)
);

CREATE TABLE Payments (
payment_id INT PRIMARY KEY AUTO_INCREMENT,
trip_id INT,
amount DECIMAL(10,2),
payment_method VARCHAR(50),
payment_time DATETIME,
FOREIGN KEY (trip_id) REFERENCES Trips(trip_id)
);
```

SQL Code to Create Tables

```
CREATE TABLE Ratings (
rating_id INT PRIMARY KEY AUTO_INCREMENT,
trip_id INT,
rating INT,
comment TEXT,
FOREIGN KEY (trip_id) REFERENCES Trips(trip_id)
);

CREATE TABLE Staff (
staff_id INT PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(100),
role VARCHAR(50),
shift_schedule VARCHAR(100)
);

CREATE TABLE Maintenance (
maintenance_id INT PRIMARY KEY AUTO_INCREMENT,
scooter_id INT,
issue_description TEXT,
technician_id INT,
status VARCHAR(20),
FOREIGN KEY (scooter_id) REFERENCES Scooters(scooter_id),
FOREIGN KEY (technician_id) REFERENCES Staff(staff_id)
);

CREATE TABLE Location_Zones (
zone_id INT PRIMARY KEY AUTO_INCREMENT,
zone_name VARCHAR(100),
gps_coordinates VARCHAR(100),
capacity INT
);
```